ME 351 Feedback Control Systems

Root Locus Method

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Idea

The *root locus* (Evans) shows how the **closed-loop poles** move as a real gain K varies from

$$\boxed{0 \leq K < \infty}.$$

For a unity negative-feedback loop, let $L(s)$ denote the complete loop transfer function, including the adjustable gain. Separate the gain from the fixed loop dynamics:

$$\boxed{L(s) = KL_0(s), \qquad K \geq 0,}$$

where $L_0(s)$ does not depend on K. The closed-loop characteristic equation $1 + L(s) = 0$ then takes *Evans form*

$$\boxed{1 + KL_0(s) = 0, \qquad L_0(s) = \frac{n(s)}{d(s)}.} \tag{1}$$

Equivalently, $d(s) + Kn(s) = 0$. We consider proper rational $L_0(s)$, with pole count $n \geq m$ (zero count) after cancellations.

The root locus is the set of all closed-loop poles s satisfying this characteristic equation as K varies.

Magnitude and Phase Conditions

The closed-loop poles s are in general complex. Therefore, at a candidate pole location s, the characteristic equation

$$KL_0(s) = -1$$

is a complex equation. We choose to represent this equation in polar form, so equality requires both magnitude and phase to match.

Since

$$-1 = e^{j(2\ell+1)\pi}, \qquad \ell \in \mathbb{Z},$$

we obtain the *magnitude condition*

$$\boxed{K|L_0(s)| = 1} \tag{2}$$

and the *phase condition*

$$\boxed{\angle L_0(s) = (2\ell + 1)\pi.} \tag{3}$$

Thus, the two conditions have distinct roles:

phase condition	$\longrightarrow$	which locations s can be closed-loop poles?
magnitude condition	$\longrightarrow$	what gain K places a pole at a chosen location $s^\star$?

To interpret the phase condition geometrically, factor the loop transfer function into its poles p_i and zeros z_j:

$$\boxed{L_0(s) = \frac{n(s)}{d(s)} = k_0 \frac{\prod_{j=1}^{m}(s - z_j)}{\prod_{i=1}^{n}(s - p_i)}.} \tag{4}$$

To identify k_0, write

$$n(s) = b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0 = b_m \prod_{j=1}^{m}(s - z_j),$$

and

$$d(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0 = a_n \prod_{i=1}^{n}(s - p_i).$$

Hence,

$$\boxed{k_0 = \frac{b_m}{a_n}.} \tag{5}$$

Thus, k_0 is simply the ratio of the leading coefficients of $n(s)$ and $d(s)$.

The phase condition becomes

$$\boxed{\angle k_0 + \sum_{j=1}^{m} \angle(s - z_j) - \sum_{i=1}^{n} \angle(s - p_i) = (2\ell + 1)\pi.} \tag{6}$$

For $k_0 > 0$, $\angle k_0 = 0$, so

$$\boxed{\sum_{j=1}^{m} \angle(s - z_j) - \sum_{i=1}^{n} \angle(s - p_i) = (2\ell + 1)\pi.} \tag{7}$$

For $k_0 < 0$, the additional phase of π changes the corresponding shortcut rules; these are summarized later and derived as an exercise.

Minimal Root-Locus Rules

Assume $k_0 > 0$, with n open-loop poles and m finite open-loop zeros.

There are n root-locus branches.

0. Symmetry

For systems with real coefficients,

$$\boxed{\text{The root locus is symmetric about the real axis.}} \tag{8}$$

1. Start and End Points

The characteristic equation can be written

$$d(s) + Kn(s) = 0.$$

At $K = 0$,

$$d(s) = 0.$$

Therefore, every branch starts at an open-loop pole:

$$\boxed{K = 0 : \qquad s = p_i.} \tag{9}$$

As $K \to \infty$, divide the characteristic equation by K:

$$\frac{1}{K} d(s) + n(s) = 0.$$

For finite s,

$$n(s) = 0.$$

Therefore, branches end at open-loop zeros:

$$\boxed{K \to \infty : \qquad s \to z_j.} \tag{10}$$

If there are fewer zeros than poles, only m branches can end at finite zeros. The remaining

$$\boxed{r = n - m} \tag{11}$$

branches go to infinity.

The integer r is the *relative degree*.

Thus,

branches start at open-loop poles and end at open-loop zeros or at infinity.

2. Real-Axis Segments

Consider a candidate point s on the real axis. The phase condition is

$$\sum_j \angle(s - z_j) - \sum_i \angle(s - p_i) = (2\ell + 1)\pi.$$

For a real pole or zero, the angle seen from s is particularly simple. A pole or zero to the left contributes zero phase. A zero to the right contributes $+\pi$, while a pole to the right contributes $-\pi$.

Modulo 2π,

$$-\pi \equiv \pi,$$

so every real pole or zero to the right contributes π to the phase condition. Thus, if N_R is the total number of real poles and zeros to the right of s,

$$\angle L_0(s) = N_R\pi \qquad (\text{mod } 2\pi).$$

The Evans phase condition requires an odd multiple of π. Therefore,

$$N_R = 1, 3, 5, \ldots$$

and hence

$$\boxed{\text{The real-axis locus has an odd number of real poles and zeros to its right.}} \tag{12}$$

A complex-conjugate pair of poles or zeros contributes zero net phase modulo 2π on the real axis; only real poles and zeros are counted.

The real-axis rule is therefore simply the Evans phase condition specialized to the real axis.

3. Asymptotes: What Happens at Infinity?

Suppose

$$r = n - m > 0.$$

Then r branches must go to infinity.

For $|s|$ much larger than the magnitude of every finite pole and zero,

$$s - p_i \approx s, \qquad s - z_j \approx s.$$

Therefore,

$$\boxed{L_0(s) \sim \frac{k_0}{s^r}.} \tag{13}$$

Thus, far from all finite poles and zeros, the detailed pole–zero locations become unimportant. The geometry is determined by the *excess number of poles over zeros.*

At large $|s|$, relative degree determines the root-locus geometry.

Asymptote angles

Write

$$s = \rho e^{j\theta}.$$

Since $k_0 > 0$,

$$L_0(s) \sim \frac{k_0}{\rho^r} e^{-jr\theta},$$

and therefore

$$\angle L_0(s) \sim -r\theta.$$

The Evans phase condition requires

$$-r\theta = (2\ell+1)\pi \qquad (\text{mod } 2\pi).$$

Hence, the r distinct asymptotic directions are

$$\boxed{\theta_q = \frac{(2q+1)\pi}{r}, \qquad q = 0, 1, \ldots, r-1.} \tag{14}$$

Thus, the familiar asymptote-angle rule is simply the *root-locus phase condition applied as* $|s| \to \infty$.

For example, if $r = 3$,

$$L_0(s) \sim \frac{k_0}{s^3},$$

and the three asymptotic directions are

$$\boxed{\theta = 60^\circ, \qquad 180^\circ, \qquad 300^\circ.}$$

Centroid

The approximation $L_0(s) \sim k_0/s^r$ gives the asymptote directions. To determine where the asymptotes intersect the real axis, retain one more term in the large-s behavior.

Write

$$\prod_{j=1}^{m}(s - z_j) = s^m - \left(\sum_j z_j\right) s^{m-1} + \cdots$$

and

$$\prod_{i=1}^{n}(s - p_i) = s^n - \left(\sum_i p_i\right) s^{n-1} + \cdots .$$

For large s,

$$L_0(s) \sim \frac{k_0}{s^r}\left[1 + \frac{\sum_i p_i - \sum_j z_j}{s} + \cdots\right].$$

Compare this with

$$\frac{k_0}{(s-\alpha)^r}.$$

For large s,

$$\frac{k_0}{(s-\alpha)^r} = \frac{k_0}{s^r}\left(1 - \frac{\alpha}{s}\right)^{-r} \sim \frac{k_0}{s^r}\left(1 + \frac{r\alpha}{s} + \cdots\right).$$

Matching the $1/s$ terms gives

$$r\alpha = \sum_i p_i - \sum_j z_j.$$

Therefore,

$$\boxed{\alpha = \frac{\sum_i p_i - \sum_j z_j}{r}.} \tag{15}$$

The r branches going to infinity therefore approach straight lines passing through the centroid α at the angles θ_q.

For the first few relative degrees,

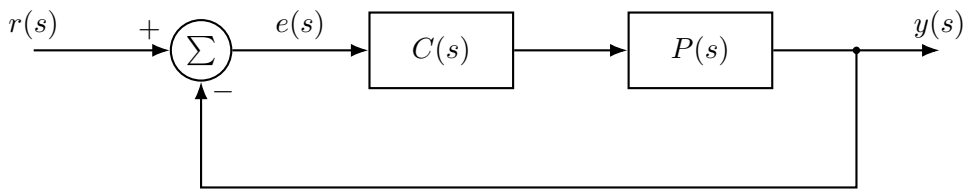
r	asymptote angles
1	180°
2	$90^\circ,\ 270^\circ$
3	$60^\circ,\ 180^\circ,\ 300^\circ$
4	$45^\circ,\ 135^\circ,\ 225^\circ,\ 315^\circ$

Worked Examples: Proportional, PD, and Actuator Lag

Consider the standard unity negative-feedback configuration shown in the Figure below. The reference is r, the tracking error is $e = r - y$, and the output is y. The closed-loop transfer function is

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)},$$

so the characteristic equation is $1 + C(s)P(s) = 0$.



In all three examples the nominal plant is

$$\boxed{P(s) = \frac{1}{s^2 + 1}},$$

with open-loop poles at $s = \pm j$. We compare the root loci for $K \geq 0$ and interpret the changes using the rules developed above.

Example 1: Proportional Control

Let $C(s) = K$. Evans form is

$$\boxed{1 + K \underbrace{\frac{1}{s^2 + 1}}_{L_0(s) = P(s)} = 0}, \qquad s^2 + (1 + K) = 0.$$

There are two open-loop poles $\pm j$, no finite zeros, and $r = 2$. The asymptotes have angles $90°$ and $270°$ and centroid $\alpha = 0$. Indeed,

$$\boxed{s_{1,2}(K) = \pm j\sqrt{1+K}}.$$

Both branches remain on the imaginary axis: proportional control increases the oscillation frequency but *does not introduce damping*. Comparison with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ gives $\omega_n = \sqrt{1+K}$ and $\zeta = 0$. The closed-loop system is not asymptotically stable for any $K \geq 0$.

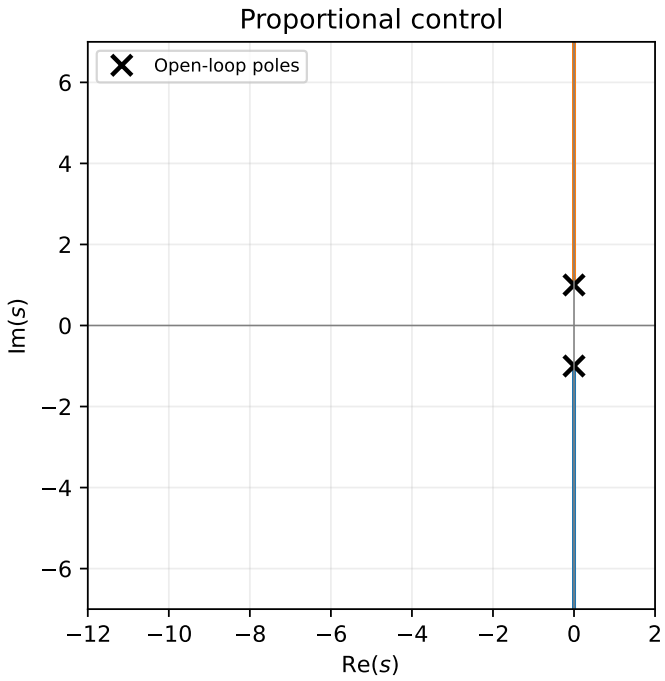


Figure 1: Proportional control. Crosses indicate open-loop poles.

Example 2: PD Control

Let $C(s) = K(1+s)$. Evans form is

$$\boxed{1 + K \underbrace{\frac{s+1}{s^2+1}}_{L_0(s)=(1+s)P(s)} = 0}, \qquad s^2 + Ks + (1+K) = 0.$$

The open-loop poles are $\pm j$, with one zero at $s = -1$. There are two branches and one asymptote ($r = 1$), directed along the negative real axis at $180°$. The centroid is not needed to sketch this single horizontal asymptote. The real-axis locus is $(-\infty, -1)$; one branch approaches -1 and the other goes to negative infinity. For $K > 0$, all coefficients of the second-order characteristic polynomial are positive, so both poles lie in the open left half-plane. In particular, PD action introduces damping.

The poles are

$$\boxed{s_{1,2}(K) = \frac{-K \pm \sqrt{K^2 - 4(1+K)}}{2}}.$$

They meet at $K_b = 2 + 2\sqrt{2}$ at the real-axis point $\boxed{s_b = -1 - \sqrt{2}}$, then separate toward -1 and $-\infty$.

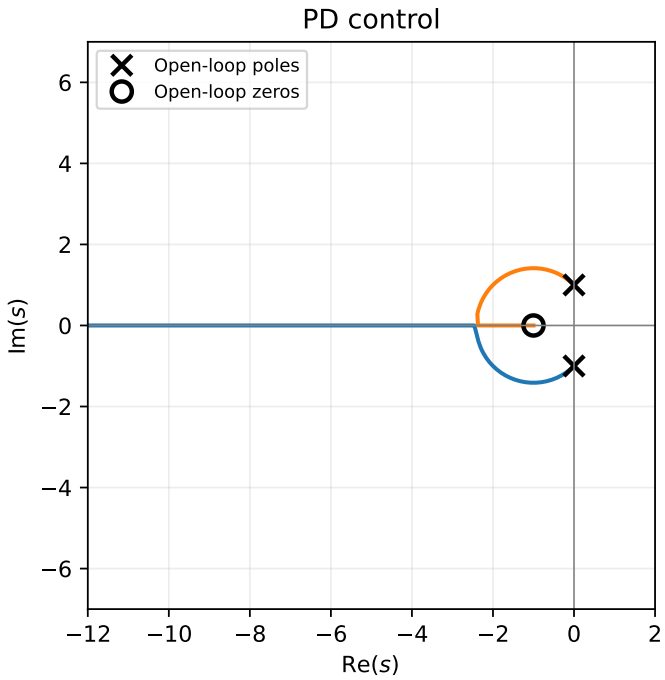


Figure 2: PD control adds a zero at -1, bending the locus into the left half-plane.

Example 3: Proportional Control with an Actuator Lag

Model the additional first-order dynamics by

$$D(s) = \frac{1}{0.1s + 1}, \qquad P_{\text{new}}(s) = P(s)D(s) = \frac{1}{(s^2 + 1)(0.1s + 1)}.$$

With $C(s) = K$, Evans form is

$$\boxed{1 + K \underbrace{\frac{1}{(s^2 + 1)(0.1s + 1)}}_{L_0(s) = P_{\text{new}}(s)} = 0},$$

which gives

$$\boxed{0.1s^3 + s^2 + 0.1s + (1 + K) = 0}.$$

The open-loop poles are $\pm j$ and -10, with no zeros. There are three branches, $r = 3$, and the asymptote angles are $60°$, $180°$, and $300°$, with centroid

$$\alpha = \frac{-10 + j - j}{3} = -\frac{10}{3}.$$

The real-axis locus is $(-\infty, -10)$. The branch from -10 moves left, while the branches starting at $\pm j$ enter the *right* half-plane as K increases from zero.

Modeling note: $D(s)$ represents a first-order lag, not an exact time delay e^{-sT}. Both have the same first-order expansion near $s = 0$ when $T = \tau$, but their higher-frequency behavior differs.

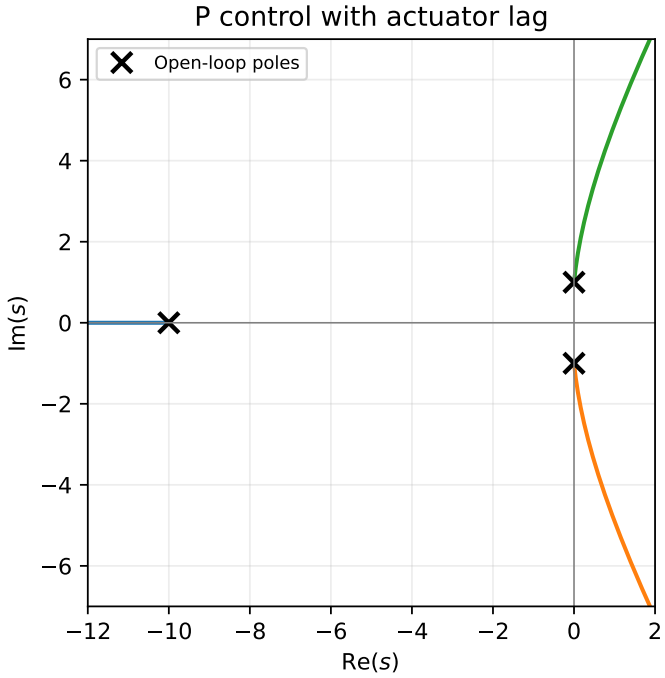


Figure 3: Additional lag introduces a pole at -10 and drives the oscillatory poles into the right half-plane.

Comparison and Interpretation

Controller / model	Added feature	Consequence for $K > 0$
P control	No added poles or zeros	Poles stay on the imaginary axis; no damping.
PD control	Zero at -1	Poles enter the left half-plane; asymptotic stability.
P with lag	Pole at -10	Oscillatory poles enter the right half-plane; instability.

Design lesson. Increasing gain does not guarantee improved stability. Zeros introduced by the controller and poles introduced by unmodeled or neglected dynamics can fundamentally change the root locus.

Root Locus as a Design Tool

The root locus reveals how the *open-loop poles and zeros* determine the possible locations of the closed-loop poles.

The distinction between changing the gain and changing the loop dynamics is fundamental:

- **Changing the gain** K moves the closed-loop poles *along an existing root locus.*
- **Changing the open-loop poles or zeros** changes the *geometry of the root locus itself.*

The three examples illustrate this distinction:

Configuration	Change in loop dynamics	Root-locus behavior
P control	No additional poles or zeros	Poles remain on the imaginary axis.
PD control	Controller zero at $s = -1$	Branches move into the left half-plane.
P with actuator lag	Additional pole at $s = -10$	Oscillatory branches move into the right half-plane.

Thus, controller design involves two complementary decisions:

Choose the loop dynamics (poles and zeros)
$\downarrow$
Shape the root locus
$\downarrow$
Choose K to locate the closed-loop poles

The closed-loop pole locations determine stability and transient behavior. For a dominant complex pole pair,

$$s = -\zeta\omega_n \pm j\omega_d,$$

the real and imaginary parts determine the decay rate and oscillation frequency, respectively.

Once a desired point $s^\star$ on the root locus has been identified, the required gain follows from the magnitude condition:

$$\boxed{K = \frac{1}{|L_0(s^\star)|}.}$$

The gain determines where we are on the root locus; the open-loop poles and zeros determine where the locus can go.

Summary

For

$$1 + KL_0(s) = 0, \qquad K \geq 0,$$

first determine the sign of

$$\boxed{k_0 = \frac{\text{leading numerator coefficient}}{\text{leading denominator coefficient}}.}$$

For $k_0 > 0$, the essential rules are

Phase:	$\sum_j \angle(s - z_j) - \sum_i \angle(s - p_i) = (2\ell + 1)\pi,$
Magnitude:	$K = \dfrac{1}{\lvert L_0(s)\rvert},$
Start:	open-loop poles,
End:	open-loop zeros or infinity,
Real axis:	odd number of real poles/zeros to the right,
Relative degree:	$r = n - m,$
Centroid:	$\alpha = \dfrac{\sum_i p_i - \sum_j z_j}{r},$
Asymptotes:	$\theta_q = \dfrac{(2q + 1)\pi}{r}.$

The construction rules are consequences of the characteristic equation and its magnitude and phase conditions, rather than independent rules to memorize.

Exercise: Negative k_0

Suppose

$$k_0 < 0.$$

Starting from

$$1 + KL_0(s) = 0, \qquad K \geq 0,$$

and accounting for the phase contributed by k_0, derive:

1. the pole–zero phase condition;
2. the real-axis rule;
3. the asymptote angles.

Verify that the results are

$$\boxed{\sum_j \angle(s - z_j) - \sum_i \angle(s - p_i) = 2\ell\pi,}$$

$$\boxed{\text{real-axis locus: even number of real poles and zeros to the right,}}$$

and

$$\boxed{\theta_q = \frac{2q\pi}{r}, \qquad q = 0, \ldots, r-1.}$$

Explain why the start/end points and centroid do not change.